

Review Study on Synthesis and Applications of Orange Peel Extract

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Abstract

Orange refers to Citrus sinensis, also called sweet orange. Botanically classified as berries, having bright (green to yellow) coloured skin. Outer rind or skin known as epicarp (or flavedo). Orange peels are the byproducts remaining after consumption of oranges. The conventional extraction technique used for preparation of various orange peel extract are powder extract, aqueous extract, solvent extract and Soxhlet extraction. For extracting essential oils from citrus peels hydodistillation and cold pressing method is used. Nowadays, many innovative techniques are used for extraction of essential oils. Qualitative analysis of orange peel extract shows various plant secondary metabolites such as saponins, anthracene glycosides, reducing sugar, cardiac glycosides, flavonoids, tannins, alkaloids, phenols and terpenoids. Orange peels are effective in the management of arthritis, cancer, typhoid, DNA damage, heart disease, inflammation, blood clot, stomach ulcers, arteriosclerosis, kidney stones and blood pressure.

Keywords

Orange Peel, Extract, Biopesticides, Insecticides.

Introduction

Oranges are the most widely used citrus fruit for its nutritional and medicinal values. An orange is a fruit comes under citrus genus in the family Rutaceae and it primarily refers to Citrus sinensis, which is also called sweet orange. It is originated from tropical and sub-tropical regions of the world. The fruits are botanically classified as berries and could be termed as a hesperidium. Citrus fruits have rough, robust and bright (green to yellow) coloured skin. They are usually 4 to 30 cm long and 4 to 20 cm in diameter, with a leathery surrounding rind or skin known as epicarp (or flavedo) that covers the fruits and protects it from damages (Oby and Chinonyerem, 2019). India produces around 25 lakh tonnes of orange every year. Consumption of orange fruits generates orange peel wastes which are rich in nutrients and contain many phytochemicals which are useful in making drugs and food items (Gotmare and Gade, 2018).

Objective of study

The aim of this review study was to explore the synthesis methods of orange peel extract as well as its applications.

Review of Literature

Processing of orange peels into potential resource helps in indirect waste management which ultimately promotes a safer and healthier environment. Report shows that essential oils of some citrus species have insecticidal properties against insect pests (Badawy, 2015). Limonene's 1% mixture was used to control mealybugs and scale insects and safe for most plants in this proportion. 69 to 100% of mealybugs and scale Insects are controlled by full strength mixture (Zewde and Jembere, 2010). Orange peel contains many phytochemicals like vitamins, flavonoids, lignin's, carotenoids, saponin, plant sterol and terpenoids which shows antioxidant property. Ascorbic acid, flavonoids are the main compounds in citrus peel which are good for human health. The flavonoids in orange peel species are of mainly three types - Eriocitrin, Narirutin, Hesperidin and Naringin. Citrus peel also contains 1.5% phyto constituent essential oil, vitamin-C, D- limonene, lemon oil & pectins (Parmar et al., 2020).

Orange fruit contains flavonoids, alkaloids, limonoids group and coumarins while synephrine, flavonones, auraptene and limonin are the most dominant bioactive component (Parmar et al., 2020). Orange peel oil have toxicity towards weevils, Callosobruchus maculatus and Culex pipiens. Fumigation activity was also reported against fleas and household insects like Musca domestica, Blatella germanica, fleas and Sitophilus oryzae (Zewde and Jembere, 2010). Valuable compounds are present in the byproduct after the processing of citrus fruits. Oranges singly contributes to > 60% of the

citrus fruit production in the world. Most flavours of commercial values are found in the genus citrus and the subgenus Eucitrus (Baker et al., 2022). Cow pea treated with the powder of orange peels is associated with LD50 of 4% (w/w) for *Callosobruchus maculatus* (F.) exposed to it (Don Pedro, 1985).

Methodology

1. Methods of Orange Peel Extract Synthesis

1.1. Preparation of Powder

The fresh oranges were washed with tap water. The peel was separated and cut into small pieces followed by shade or sun drying for 92 hrs or in oven for a period of 6-7 days at a temperature of 30°C. Dried peels were ground to make coarse powder by using pestle and mortar and stored in zip lock bags or in air tight bottles for future use. [Gotmare and Grade, 2018, Arora and Kaur 2013, Baker et al 2022, Oboh et al. 2017].

Fresh orange peel (flavado) were dried in two different ways. One part was shade dried at 30-33°C for 7 days and then ground to fine powder while other part was grounded immediately after peeling the oranges by using mortar and pestle (Oikeh et al 2020). Fine powder of orange peel is obtained by using electric blender. Then the powder was dried at 40°C temperature for 24 hr in an oven (Abdelazem et al., 2021). Orange peels were air dried under room temperature for 14 days (Gariba et al., 2021). Orange peel pieces were air dried for 21-28 days and then grounded into fine powder by using electric grinder (Ojebode et al., 2016).

1.2. Preparation of Aqueous Extract

Orange peel powder was soaked in distilled water under shaking condition at room temperature for 24 hr. The obtained extract was then filtered by using Whatman filter paper No. 1 then concentrated to dryness with the help of water bath at 70°C. Extract was kept at 4°C in glass vials for future use (Arora and Kaur 2013). The filtered extract and water bath were used for concentrating the orange peel extract. Water bath maintained at 75°C constantly (Gotmare and Gade, 2018).

Orange peel powder mixed with water then mixture was heated at 60°C for 45 minutes under water bath then the mixture was filtered by using Whatman filter paper overnight. Obtained filtrate was centrifuged at 2500 rpm for 20 min. Then the collected supernatant was stored at 4°C in Falcon tube for future use (Ahamed et al., 2017). Dried rind and aril of orange peel were soaked in distilled water separately for 24 hours followed by filtration (Abdelazem et al., 2021).

1.3. Preparation of Solvent Extract

Fresh orange peels were chopped with a knife and soaked in acetone, distilled water, petroleum ether and ethanol at different rates. After 24 hours the mixture was filtered with cheese cloth and by using Whatman No. 9 filter paper (Zewede and Jembere, 2010). Powdered orange peel was taken and added to 80% ethanol. Then the mixture was kept for 72 hours at room temperature under the shaking condition by using rotatory shaker. Then the mixture was filtered by using Whatman filter paper & the filtrate was evaporated at 55°C in evaporator. After that collected concentrates after complete evaporation was used to make 1% solution by using 0.9% NaCl solution and kept at 4°C for future use (Ahamed et al., 2017).

Orange peel powder was treated with 90% ethanol for 24 hrs at room temperature (Chayengia et al., 2010). Orange peel powder was added into 100% methanol. The mixture was covered with Para film and kept for 48 hr in a shaker. The solution was filtered and concentrated at 60°C by using rotary evaporator after which residues were dissolved in acetone. The final solution kept in a refrigerator (Gariba et al., 2021). Orange peel powder was soaked in n-hexane on Magnetic stirrer. The extracts were filtered by using filter paper & muslin cloth after 72 hr. The filtrates were concentrated and then dried in an oven at 4°C and 600 mm Hg for solvent reduction. (Ojebode et al., 2016). Dried rind and aril of orange peel were soaked in chloroform and ether separately for 24 hrs after that filtered the mixture. (Abdelazem et al., 2021).

1.4. Soxhlet Extraction

Soxhlet extraction method was used when the solubility of desired compound is limited in solvent and the impurity is insoluble in that solvent. It recycles a very small amount of solvent efficiently by dissolving a larger amount of material. (Oby and Chinonyerem, 2019). The orange peel powder will be extracted with different solvents: hexane, methanol, acetone by Soxhlet Extractor. Orange peel powder used for extraction with solvent (hexane / methanol / acetone) at 50°C for 5 hrs by Soxhlet extraction method.

After the extraction the extract filtered through a Whatman No. 2 filter paper for removal of any peel particles present in extract. The filtered extract then evaporated to dryness under vacuum at 60°C by a rotary evaporator. The extracts will be stored in refrigerator at 4°C for further use. (Gotmare and Gade, 2018).

Dry the orange peel at 41°C in an oven for 48 hrs. Dried peel converted to fine powder. Add ethanol, methanol for 6 hrs at room temperature by soxhlet method. For complete extraction the residues was re-extracted 2-3 times. The solution was filtered and the extract was evaporated at low pressure at 60° C by using rotary evaporator (Parmar et al., 2020).

1.5. Extraction of Essential oil from orange peels

Essential oils from orange peels can be obtained by various methods. Hydrodistillation involves the heating a mixture of water or other solvent and orange peel to obtain essential oil by evaporation followed by liquefaction of the vapours in condenser. Steam distillation uses the steam which is percolating the orange peels to obtain the essential oil. Cold Pressing method involves oil separation by pressing method at low temperature and pressure. Nowadays, for extraction of essential oil many innovative techniques are used that is safer, efficient and energy saving.

- i. Ultrasound assisted extraction (UAE)
- ii. Microwave assisted extraction (MAE)
- iii. Supercritical fluid extraction (SFE)
- iv. Accelerated Solvent extraction (ASE) (Oby and Chinonyerem, 2019)

Result and Discussion

2. Biochemical Components of Orange Peel Extract

Orange peel has many phytochemicals due to which it shows antioxidant property and includes vitamins, flavonoids, lignins, carotenoids, saponin, plant sterol, terpenoids. Phytochemicals are non-nutritive in nature, naturally present in plants and having a potential in curing many diseases. Around 4000 phytochemicals are found to be proved by researchers (Mercy, 2015). Based on the basic chemical structure and its properties phytochemicals are grouped under the categories - Terpenoids, Alkaloids, Anthocyanins, Lignans, Flavonoids, Phytosterol, Polyphenol, Saponin etc. (Nanna et al., 2013). Orange peel remained after processing is a useful byproduct which are good sources of bioactive compounds showed by the research. (Mamta and Parminder, 2013).

Ascorbic acid and flavonoid are the main compounds in citrus peel which are beneficial for the human health. Eriocitrin, Narirutin, Hesperidin and Naringin are the main types of flavonoids occur in orange peel species. Lemon oil, D-limonene, Vitamin-C & pectins are found in citrus peel and also 1.5% phyto-constituent essential oil also present. Flavonoids, alkaloids, limonoids groups and coumarins are the compounds present in orange peel while synephrine, flavonones, limonene and auraptene are the most dominant compound in orange peel (Parmar et al., 2020). Citrus sinensis peels extract's qualitative assay was carried out on plant secondary metabolites such as saponins, anthracene glycosides, reducing sugar, cardiac glycosides, flavonoids, tannins & alkaloids with standard procedure (Harbou, 1973; Trease and Evans, 2003). Higher concentrations of natural bioactive compounds such as quercetin derivatives, phenolic acids, carotenoids and saponins are found in peels towards the flesh (Goulas and Manganaris, 2012).

3. Applications of Orange Peel Extract

Minerals, vitamins and antioxidant such as flavonoids, phenolic acids, anthocyanins and carotenoids as well as many nutrients such as fiber present in oranges which boosts health immunity such as stomach and digestive system (Kumari & Kumari, 2020). These compounds have property of anti-bacterial, anti-cancer and anti-infection (Rafiq et al., 2018). Due to antimicrobial property of orange peel extract holds significance in therapeutic treatments (Lobo et al., 2010). Polyphenols act as antioxidant, antitumor and antimicrobial agent due to the free natural radical defence property which are beneficial for human health (Ighodaro, 2012). Orange peel can be used for making drugs & food items due to the presence of nutrients & many phytochemicals (Hegazy and Ibrahim, 2012). Flavonoids in oranges can suppress the cancer through anti-proliferative actions, selective cytotoxicity and by apoptosis (Elangoran et al., 1994; Hirano et al., 1994). Flavonoids like citrinsine, citricidone & saponins shows anti-typhoid activity (Khandla et al., 2020).

Flavonoids have anti- mutagenic property, having the ability to absorb Ultraviolet light and protect the DNA from damage (Stapleton and Walbot,1994). Soluble & insoluble dietary fibers act as phytochemical antioxidant found in orange fruit helps in reducing the risk of obesity, coronary heart diseases, many chronic diseases like arthritis and cancer. (Crowell, 1999) Orange flavonoid's shows inhibitory effect proliferation and on cell tumoral development by rat malignant cells in hepatic tissue of syngenetic rats and in cardiac tissue have been shown by many reports (Bracke et al.,1989). Flavonoids acts directly with the tumoral agent so to protect the DNA from damage such as in the induced chromosomal by bleomycin (Heo et al., 1994). Around 170 phyto-constituent and over 60 flavonoids present in single orange which shows anti-tumor, blood clot inhibiting, anti-inflammatory and antioxidant properties which ultimately promotes good health. (Cha et al., 2001).

Vitamin C, phytochemical compound such as limonoids, hesperdin, flavonoid, polyphenols, pectin etc. have health benefits which are useful in the treatment of stomach ulcers, arteriosclerosis, reduction in cholesterol level, prevention of cancer, kidney stones, strengthening of immune system and high Blood Pressure (Etebu and Nwauzoma, 2014). Orange peels are natural source of pectin which helps in lowering the blood sugar of post-meal rise which benefits the diabetic patients. Hesperidin and Naringin have antidiabetic property. Flavonoids impede RLIP76 proteins present in orange peel which is related to cancer & obesity. Saponins show larvicidal property. Biochemical Dlimonere present in orange peels helps in dissolving kidney stones & support Normal peristalsis by neutralizing gastric acid. Limonene (84.2%), linalool (4.4%) and Myrcene (4.1%) have anti-fungal activity (Khandla et al.,2020). Phytoconstituents like zeaxanthin, carotenoids and Beta- cryptoxanthin shows the property of reducing the risk of rheumatoid arthritis. Rheumatoid arthritis shows 52% less probability of occurence by increasing the amount of consumption of cryptoxanthin and Zeaxanthin (Milind and Dev, 2012).

Conclusion

Based on review it is concluded that orange peels contain various plant secondary metabolites such as saponins, anthracene glycosides, flavonoids, tannins, alkaloids , phenols and terpenoids. These secondary metabolites have several applications in the treatment of different diseases in humans. These biochemical components can be extracted from orange peel by conventional (powder, aqueous, solvent and soxhlet) method and advanced techniques. Essential oils from orange peels extracted by hydrodistillation and cold pressing method.

Consent

It is not applicable.

Ethical Approval

It is not applicable.

Competing Interests

Authors have declared that no competing interests exist.

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Author's Contribution

This work was carried out in collaboration of both authors. The author SG designed and analysed the work. Author PS interpreted the work as well as prepared the manuscript.

References

1. Abdelazem, R. E., Hefnawy, H. T., & El-Shorbagy, G. A. (2021). Chemical composition and phytochemical screening of Citrus sinensis (orange) peels. Zagazig Journal of Agricultural Research, 48(3): 793-804.
2. Abou Baker, D. H., Ibrahim, B. M., Abdel-Latif, Y., Hassan, N. S., Hassan, E. M., & El Gengaihi, S. (2022). Biochemical and pharmacological prospects of Citrus sinensis peel. Heliyon, 8(8): e09979.
3. Ahamed, T., Rahman, S. M., & Shohael, A. M. (2017). Thin layer chromatographic profiling and phytochemical screening of six medicinal plants in Bangladesh. International Journal of Biosciences, 11(1): 131-140.
4. Arora, M., & Kaur, P. (2013). Phytochemical screening of orange peel and pulp. International Journal of Research in Engineering and Technology, 2(12): 517-522.

5. Bracke, M. E., Vyncke, B. M., Van Larebeke, N. A., Bruyneel, E. A., De Bruyne, G. K., De Pestel, G. H., ... & Mareel, M. M. (1989). The flavonoid tangeretin inhibits invasion of MO 4 mouse cells into embryonic chick heartin vitro. *Clinical & experimental metastasis*, 7(3): 283-300.
6. Cha, J. Y., Cho, Y. S., Kim, I., Anno, T., Rahman, S. M., & Yanagita, T. (2001). Effect of hesperetin, a citrus flavonoid, on the liver triacylglycerol content and phosphatidate phosphohydrolase activity in orotic acid-fed rats. *Plant Foods for Human Nutrition*, 56(4):349-358.
7. Chayengia, B., Patgiri, P., Rahman, Z., & Sarma, S. (2010). Efficacy of different plant products against *Sitophilus oryzae* (Linn.) (Coleoptera: Curculionidae) infestation on stored rice. *Journal of Biopesticides*, 3(3): 604.
8. Crowell, P. L. (1999). Prevention and therapy of cancer by dietary monoterpenes. *The Journal of nutrition*, 129(3): 775S-778S.
9. Don-Pedro, K. N. (1985). Toxicity of some citrus peels to *Dermestes maculates* Deg. and *Callosobruchus maculatus* (F). *Journal of Stored Products Research*, 21(1):31-34.
10. Elangovan, V., Sekar, N., & Govindasamy, S. (1994). Chemopreventive potential of dietary bioflavonoids against 20-methylcholanthrene-induced tumorigenesis. *Cancer letters*, 87(1):107-113.
11. EL-BADAWY, S. S. INSECTICIDAL AND REPLENISHING ACTIVITIES OF CITRUS PEEL OILS AGAINST MEALYBUG *ICERYA SEYCHELLARUM* (WESTWOD). *Egyptian Journal of Agricultural Research*, 93(3): 791-808.
12. Etebu, E., & Nwauzoma, A. B. (2014). A review on sweet orange (*Citrus sinensis* L Osbeck): health, diseases and management. *American Journal of Research Communication*, 2(2): 33-70.
13. Gariba, S. Y., Dzidzienyo, D. K., & Eziah, V. Y. (2021). Assessment of four plant extracts as maize seed protectants against *Sitophilus zeamais* and *Prostephanus truncatus* in Ghana. *Cogent Food & Agriculture*, 7(1): 1918426.
14. Gotmare, S., & Gade, J. (2018). Orange peel: A potential source of phytochemical compounds. *Int. J. ChemTech Res*, 11(2): 240-243.
15. Goulas, V. and G. Manganaris (2012). Exploring the phytochemical content and the antioxidant potential of Citrus fruits grown in Cyprus. *Food Chem.*, 131:39-47.
16. Harborne, J. B. (1998). *Textbook of phytochemical methods. A guide to modern techniques of plant analysis*. Chapman and Hall, London, Edition, 3:135-203.
17. Hegazy, A.E. and M.I. Ibrahim (2012). Antioxidant activities of orange peel Extracts. *World Appl. Sci. J.*, 18: 684-688.
18. Heo, M. Y., Lee, S. J., Kwon, C. H., Kim, S. W., Sohn, D. H., & Au, W. W. (1994). Anticlastogenic effects of galangin against bleomycin-induced chromosomal aberrations in mouse spleen lymphocytes. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis*, 311(2): 225-229.
19. Ighodaro, O. (2012). Evaluation study on Nigerian species of *Musa paradisiaca* peels: phytochemical screening, proximate analysis, mineral composition and antimicrobial activities. *Researcher*, 4(8):17-20.
20. Khandla, S. N., Vala, M. S. & Maitreya, B. B. (2020). A REVIEW: PHYTOCHEMICAL CONSTITUENTS, MEDICINAL USES AND ECONOMICAL VALUE OF ORANGE PEELS (*Citrus sinensis* L Osbeck). *Int. J. Creat. Res. Thoughts*, 8(5) :1340-1354
21. Kumari, R. and Kumari, N. (2018). Studies on the Utilization of Orange Peel Powder in the Development of Food Product. *International Journal of Science and Research*, 9(10): 118-124.
22. Lobo, V., A. Patil, A. Phatak and N. Chandra (2010). Free radicals, antioxidants and functional foods: Impact on human health. *Pharm. Rev.*, 4: 118-126.

23. Mamta, A. and P. Kaur (2003). Phytochemical screening of orange peel and pulp. Int. J. Res. Eng. and Technol., 2 (12): 517-522.
24. Mercy, T.A. (2015). Qualitative and quantitative analysis of the phytochemicals in some selected species of the genus *Phyllanthus* in Zaria, Nigeria, 710-711.
25. Milind, P., & Dev, C. (2012). Orange: range of benefits. Int Res J Pharm, 3(7):59-63.
26. Nanna, R.S., M. Banala, A. Pamulaparthi, A. Kurra and S. Kagithoju (2013). Evaluation of Phytochemicals and Fluorescent Analysis of Seed and Leaf Extracts of *Cajanus cajan* L. Int. J. Pharm. Sci. Rev. and Res., 22 (1): 11-18.
27. Oboh, G., Ademosun, A. O., Olumuyiwa, T. A., Olasehinde, T. A., Ademiluyi, A. O., & Adeyemo, A. C. (2017). Insecticidal activity of essential oil from orange peels (*Citrus sinensis*) against *Tribolium confusum*, *Callosobruchus maculatus* and *Sitophilus oryzae* and its inhibitory effects on acetylcholinesterase and Na⁺/K⁺-ATPase activities. *Phytoparasitica*, 45: 501-508.
28. Oikeh, E. I., Oviasogie, F. E., & Omoregie, E. S. (2020). Quantitative phytochemical analysis and antimicrobial activities of fresh and dry ethanol extracts of *Citrus sinensis* (L.) Osbeck (sweet Orange) peels. *Clinical Phytoscience*, 6(1): 1-6.
29. Ojebode, M. E., Olaiya, C. O., Adegbite, A. E., Karigidi, K. O., & Ale, T. O. (2016). Efficacy of some plant extracts as storage protectants against *Callosobruchus maculatus*. *Journal of Biotechnology and Biomaterials*, 6(1):1-4.
30. Onyia, C., & Okorie, C. (2019). BIO-INSECTICIDES/REPELLENT FROM CITRUS PEELS EXTRACT AND ITS EFFICACY ON MOSQUITOES. *International Research Journal of Applied Sciences, Engineering and Technology*, 5(11): 11-20.
31. Parmar, D., Sharma, D., Pant, M., & Dan, S. (2020). Phytochemical composition and in vitro antioxidant activities of the genus *Citrus* peel extracts: a systematic review. *Int Res J Moder Eng Tech Sci*, 2(9): 953-961.
32. Shafiya, R., Rajkumari, K., Sofi, S. A., & Nadia, B. (2019). Citrus peel as a source of functional ingredient. A Review of *Journal of the Saudi Society of Agriculture science*, 7(2): 355.
34. Stapleton, A. E., & Walbot, V. (1994). Flavonoids can protect maize DNA from the induction of ultraviolet radiation damage. *Plant physiology*, 105(3): 881-889.
35. Zewde, D. K., & Jembere, B. (2010). Evaluation of orange peel *Citrus sinensis* (L) as a source of repellent, toxicant and protectant against *Zabrotes subfasciatus* (Coleoptera: bruchidae). *Momona Ethiopian Journal of Science*, 2(1): 61-75.